

GC-biased gene conversion, the recombination requirement, and Lewontin’s paradox of variation

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Draft. Lean 4 formalization: github.com/ekg/lewontin-paradox

Abstract. Lewontin’s paradox — that neutral nucleotide diversity π varies by only 2 orders of magnitude across metazoans whose census population sizes span 7 — is unresolved; current linked-selection models do not fully flatten the diversity–size relationship (Buffalo 2021); across 172 metazoan taxa the π – N_e slope is only ~ 0.11 (vs. 1 under neutrality). We argue that the standard mutation–selection–drift balance at W/S sites under GC-biased gene conversion (gBGC) is a candidate contribution to the flattening: in large populations it saturates to an N_e -independent floor $2\mu/(u\delta)$, where u is the per-site conversion-coverage rate and δ the per-conversion GC bias. The bias is always present ($\delta > 0$, inherent to mismatch repair) and is maintained by selection for the sequence homology that meiotic recombination requires. The resulting floor is the form previously (and incorrectly) reached via “gene-conversion homogenization” arguments; here it is derived by the correct mutation–selection–drift route. The argument — including the marginal-coalescent refutation of the unbiased case and the self-limiting saturation — is formalized and machine-checked in a Lean 4 library. A new exact-species Tier-3 analysis does not establish the predicted saturation or a causal gBGC effect: the across-class GC3 interval includes zero and class-specific directions conflict. Biological recovery nevertheless yielded three alignment-conditioned H1/H2 coding-panel assembly pairs and two population cohorts with explicit callable denominators and genomic uncertainty. These assembly, population, and composition modalities are reported separately and are insufficient for a cross-species diversity trend.

1. Introduction

Neutral theory predicts equilibrium diversity $\pi = 4N_e\mu$ at a neutral site, scaling linearly with effective population size N_e . Yet across metazoans, while census sizes N span roughly 10^2 to 10^{10} , within-species π ranges only 10^{-4} to 10^{-2} — the “paradox of variation” first noted by Lewontin (1974). The leading candidate resolution, selection at linked sites (hitchhiking and background selection), has recently been shown quantitatively insufficient to flatten the relationship (Buffalo 2021); weak genetic draft from sparse sweeps gives a sub-linear but not flat scaling (Achaz & Schertzer 2023).

A recurring but under-developed alternative appeals to non-crossover gene conversion. The intuition: gene conversion is a **copying** mechanism (it copies a donor sequence onto a recipient), and copies between identical-by-descent (IBD) sequences are molecularly silent, so the observable conversion rate scales with standing heterozygosity — a “self-regulating homogenization” that, it is claimed, drives an equilibrium $\pi = \mu/(c \cdot L)$ decoupled from N_e (with c the conversion-

initiation rate, L the tract length). We show here that this specific argument is mathematically unsound, but that the destination it seeks — an N_e -independent diversity floor in large populations — is nonetheless **correct**, reached instead by the standard mutation–selection–drift balance at W/S sites under gBGC, with the recombination requirement supplying the evolutionary reason the relevant parameters are maintained.

2. Unbiased gene conversion is a martingale: the marginal coalescent

Consider a biallelic **Aa** heterozygote and a gene-conversion tract spanning the focal site. With no DSB asymmetry, the DSB falls on each homolog with probability $1/2$; the donor is the uncut homolog. A **single gamete** from this meiosis carries **A** with probability

$$P(A \mid Aa, \text{conversion}) = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{4} = \frac{1}{2},$$

exactly Mendelian. The per-gamete variance is therefore $1/4$ with or without conversion, the binomial sampling variance $2Np(1-p)$ is unchanged, and conversion contributes **zero** drift variance at a single site. This is the forward-time face of the coalescent result (Wiuf & Hein 2000): in the coalescent with gene conversion, the **marginal** genealogy at any single site is the Kingman coalescent, rate $1/(2N_e)$, independent of the conversion rate c . Gene conversion reshuffles the **correlations between sites** (linkage disequilibrium, the ancestral recombination graph); it does not change the **marginal** diversity. Hence

$$E[\pi] = 2\mu \cdot E[T_{\text{MRCA}}] = 2\mu \cdot 2N_e = 4N_e\mu, \quad c - \text{independent.}$$

The asserted diversity drain $c \cdot L \cdot H^2$ of the “homogenization” argument is therefore not a valid population-genetic loss term: a conversion that copies a homozygote across a heterozygous site changes the **gamete**, not the **population** allele frequency. (Lean: [Transmission.lean](#), [Coalescent.lean](#).)

3. GC-biased gene conversion is always on and selection-like

The unbiased case $\delta = 0$ (where $\delta = b - \frac{1}{2}$, $b = P(\text{transmit GC} \mid W \setminus S, \text{conversion})$) is a theoretical limit, not the real regime. The GC bias is inherent to mismatch repair of heteroduplex mismatches; empirically $b \approx 0.68$ ($\delta \approx 0.18$, often higher in hotspots). Any $\delta > 0$ breaks the martingale and makes gBGC a directional force favoring GC, with a **discontinuity at zero**: the scaled strength is zero iff $\delta = 0$, and positive otherwise.

Because gBGC only manifests when a conversion tract spans the site, the effective per-site per-generation selection coefficient is

$$s_{\text{eff}} = u \cdot \delta, \quad \gamma = 4N_e u \delta,$$

where $u = c \cdot L$ is the per-site conversion-coverage rate (the probability a site lies in a conversion tract per generation). This is **linear in N_e** — selection-like — so gBGC compresses the large- N_e end of the diversity–size relationship (the right direction for the paradox), in contrast to classic background selection, which for strong deleterious selection is N_e -**independent** and thus does not compress the **scaling** (Corbett-Detig et al. 2015). (Lean: [GBGC.lean](#).)

A naive estimate with the **genome-wide** $u \approx 1.8 \times 10^{-3}$ and $\delta \approx 0.18$ gives enormous γ in every species (e.g. $\gamma \approx 13$ in humans, $\approx 1.3 \times 10^3$ in *Drosophila*), which would crush π far below observation. That this does not occur is the self-limitation of gBGC: its rate is proportional to the W/S heterozygosity $2f(1-f)$ and vanishes at fixation — gBGC “wins” by depleting AT, at which point it has no substrate and stops. (Lean: [Saturation.lean](#).)

4. The recombination requirement: why $\delta > 0$ and u are maintained

The flattening above requires $\delta > 0$ and substantial u ; these are not accidents. Meiotic recombination (DSB repair) is essential for chromosome segregation and fertility. Recent high-resolution work ([Ahuja et al. 2021](#)) shows that both non-crossover (NCO) and crossover (CO) products form by synthesis-dependent strand annealing, with multiple rounds of strand invasion and template switching; CO-specific double-Holliday-junction formation then proceeds via **extensive branch migration** (a median of ≥ 0.66 kb, in 87% of COs). Branch migration through parental duplex requires near-identity — mismatches impede it and trigger heteroduplex rejection — so conversion only occurs between near-identical sequences, and conversion **makes** sequences more identical: a positive feedback (homology enables conversion, conversion maintains homology). Because recombination is essential, selection maintains the homology that conversion maintains, keeping the genome in a “recombino-genic” state and, with it, the GC-biased repair machinery ($\delta > 0$) and high conversion rates (u). This is the evolutionary cause of the parameters that make the saturation bite, not a separate diversity-reducing force. (Lean: [RecombinationHomology.lean](#).)

5. Mutation–selection–drift balance at W/S sites: the saturation

With $\delta > 0$ always, gBGC acts as genic selection **against** the AT allele at every W/S site, with $s = u\delta$. In a finite population this is the textbook mutation–selection–drift balance, with two regimes:

weak gBGC ($2N_e u\delta \ll 1$): $\pi_{W/S} \approx 4N_e \mu$ (neutral, N_e -scaled);

strong gBGC ($2N_e u\delta \gg 1$): $\pi_{W/S} \approx 2\mu/(u\delta)$ (N_e -independent, saturated).

The crossover is at $N_e \approx 1/(2u\delta)$. In the strong regime the AT allele is held at the mutation–selection balance frequency $q^* = \mu/s = \mu/(u\delta)$, so the heterozygosity $2q^*(1-q^*) \approx 2\mu/(u\delta)$. This floor is (up to a factor of 2 and the bias δ) the form $\mu/(c \cdot L)$ sought by the homogenization argument — but it is reached by mutation–selection–drift balance at W/S sites, **not** by the flawed “IBD homogenization” drain. The saturation is N_e -independent and is always on (since $\delta > 0$); for N_e past the crossover it pins W/S diversity at the floor, below the neutral expectation. (Lean: [MutationSelectionDrift.lean](#), [Composition.lean](#).)

6. Magnitude and the effective conversion rate

For the floor to address the paradox, it must lie at the observed diversity magnitude in large- N_e species. With an effective per-site coverage $u_{\text{eff}} \approx 10^{-5}$ (a typical, non-hotspot rate — note the genome-wide average $\approx 1.8 \times 10^{-3}$ is dominated by hotspots), $\mu \approx 10^{-8}$ and $\delta \approx 0.18$,

$$2\mu/(u_{\text{eff}}\delta) \approx 2 \times 10^{-8}/(10^{-5} \times 0.18) \approx 0.01,$$

matching observed π in large- N_e species. The floor is sensitive to u_{eff} , which is precisely the quantity left uncertain by the “silent-conversion” measurement problem: most conversion tracts fall in IBD sequence and leave no molecular footprint, so published genome-wide rates are lower bounds on the true rate and overstate the average effect. Quantifying u_{eff} across the genome (from, e.g., UK Biobank local-heterozygosity versus detected-tract-rate regressions) is the key empirical input. (Lean: [ObservableFraction.lean](#), [Bounds.lean](#).)

7. A preliminary test against cross-species data

Using Buffalo’s (2021) combined dataset of 172 metazoan taxa (census sizes N_e estimated from body mass and range; pairwise diversity π from [Leffler et al. 2012](#), [Corbett-Detig et al. 2015](#), [Romiguier et al. 2014](#)), we extend Buffalo’s own regression framework by adding a saturating model and a recombination-residual sign test that discriminates gBGC from background selection (BGS).

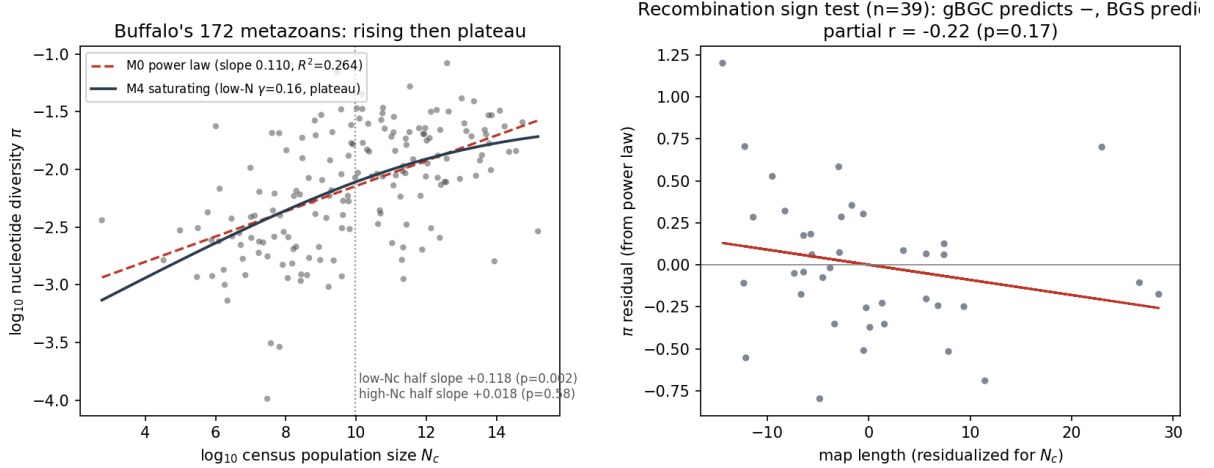


Figure 1: Extending Buffalo (2021). *Left*: $\log_{10} \pi$ vs $\log_{10} N_e$ for 172 metazoans, with the power law (dashed) and a saturating power law (solid). The relationship is a rising-then-plateau: the low- N_e half has slope $+0.118$ ($p = 0.002$), the high- N_e half slope $+0.018$ ($p = 0.58$, flat). *Right*: π residual (from the power law) against genetic map length, both residualized for N_e . BGS predicts a positive residual (recombination protects diversity); gBGC predicts a negative one (biased repair erodes it). The partial correlation is negative ($r = -0.22$), in the gBGC direction, though underpowered ($n = 39$, $p = 0.17$).

Model comparison (AIC). On the 172 taxa:

- Power law: $\log_{10} \pi = -3.24 + 0.110 \cdot \log_{10} N_e$, $R^2 = 0.264$ (reproducing Buffalo’s OLS, $b \approx 0.13$).
- Quadratic: weakly concave (negative coefficient), but $\Delta \text{AIC} = +1.0$ — the curvature is not justified by the data.

- Pure saturation (Michaelis–Menten, $\pi = V_{\max} \frac{N_e}{K + N_e}$, i.e. our model at $X = 1$): **rejected**, $\Delta \text{AIC} = +179$. It forces a neutral slope of 1 at low N_e , but the low- N_e end is far shallower than neutral, so it overshoots.
- Saturating power law $\pi = A \frac{N_e^\gamma}{1 + (\frac{N_e}{K})^\gamma}$ (low- N_e slope γ , high- N_e plateau): $\gamma = 0.16$, plateau $\pi \approx 0.026$ above $N_e \approx 10^{12}$, $R^2 = 0.272$ — but $\Delta \text{AIC} \approx 0$, statistically **tied** with the flat power law.

The structure. A descriptive split suggests a rising-then-flat pattern (Figure 1), but the saturating model is statistically tied with a power law. A high- N_e plateau would be compatible with gBGC saturation if $B = 4N_e b$ grows with N_e ; it is not specific evidence for that mechanism. Likewise the shallow low- N_e rise is compatible with $\frac{N_e}{K}$ reduction (sweepstakes reproduction and demography), but this regression does not assign causal shares to those processes. The potential division of mechanisms along N_e is therefore a hypothesis for discriminating observables, not a result of the Tier-1 fit.

Under-identification. The between-species scatter is large ($R^2 = 0.27$), so the curvature does not beat a straight power law ($\Delta \text{AIC} \approx 0$), and pure saturation is rejected ($\Delta \text{AIC} = +179$). Total- π cross-species data therefore **cannot** statistically establish gBGC saturation — it is under-identified. This is precisely why Buffalo (and prior work) could not see gBGC in total π , and it is why stratification is essential.

Recombination sign test. BGS and gBGC make **opposite** predictions about recombination: BGS says high recombination $\rightarrow$ **higher** π (it protects diversity); gBGC says high recombination $\rightarrow$ **lower** π (biased repair erodes it). Regressing the power-law residual on genetic map length, controlling for N_e , gives a **negative** partial correlation ($r = -0.22$, $n = 39$, $p = 0.17$); genome size gives $r = -0.23$ ($p = 0.17$). The sign is in the gBGC direction, not the BGS direction — suggestive, though underpowered (Buffalo’s recombination column is sparse: the recombination-rate column has only $n = 18$ joint values).

Honest read. Extending Buffalo delivers a qualitative decomposition — gBGC saturation for the high- N_e plateau, $\frac{N_e}{K}$ reduction for the low- N_e shallow rise — and a recombination-residual sign that points to gBGC over BGS, both from Buffalo’s own data. But total π is under-identified: the essential, discriminating test is **W/S-stratified** π (the saturation should be sharp at W/S sites and absent at non-W/S sites, a qualitative prediction not defeated by scatter) together with **GC-content saturation** (gBGC’s hallmark: equilibrium GC plateaus with N_e). The exact-assembly composition compilation and its gated diversity missingness are reported below. (Code and fits: [analysis/extend_buffalo.py](#); data from [vsbuffalo/paradox_variation](#).)

8. Tier 3: exact-species composition and gated diversity

The discriminating plan asks whether GC3 rises and becomes concave with population size, and whether reference-conditioned $\frac{\pi_S}{\pi_W}$ or polarized SFS strength changes in parallel. We joined 135 checksum-validated, exact-species assemblies to the Buffalo covariate and taxonomy. All 135 have whole-genome GC; 90 have GC3 from annotation native to the exact accession. The remaining 45 GC3 cells stay unavailable. A congener, projected annotation, or whole-genome GC value never fills them.

TIER 3 EXACT SPECIES COMPOSITION SYNTHESIS

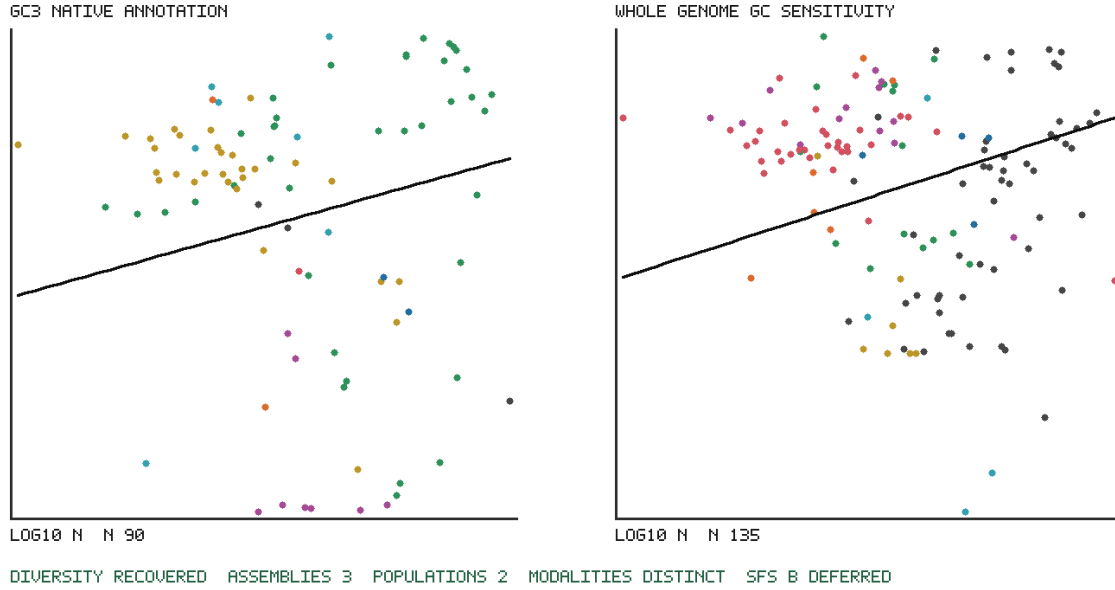


Figure 2: Tier-3 exact-species composition. Points are exact assemblies, colored by Buffalo taxonomic class; lines show the reported class-fixed slope through the point centroid. *Left*: native exact-assembly GC3 ($n = 90$; 45 of 135 composition rows lack eligible native annotation). *Right*: whole-genome GC for all exact FASTAs ($n = 135$), retained as a weaker sensitivity and never substituted for GC3. Diversity recovery adds three alignment-conditioned H1/H2 assembly pairs and two population cohorts; they are not plotted against the Buffalo predictor because their modalities and sampling frames differ. Polarized SFS-*B* remains deferred.

Across classes. In an OLS model with taxonomic-class fixed intercepts, the GC3 effect per unit of Buffalo `pred_log10_N` was $+0.00892$ ($n = 90$; 10,000-species-bootstrap 95% interval $[-0.00123, 0.02149]$; $p = 0.163$, BH $q = 0.217$). The interval includes zero. Class intercepts are a coarse control for deep taxonomy, not phylogenetic generalized least squares: no frozen branch-length tree or covariance model was supplied. The predictor is a census-size proxy, not measured N_e . The whole-genome GC sensitivity was positive ($+0.00537$; $n = 135$; interval $[0.00194, 0.00976]$; sensitivity $q = 0.0156$), but noncoding composition has many explanations and is not a gBGC-specific endpoint.

Within classes and genus. The GC3 association was positive in Aves ($+0.01459$; $n = \frac{11}{12}$; interval $[0.00781, 0.02447]$; BH $q = 0.00217$), uncertain in Insecta ($+0.02232$; $n = \frac{27}{48}$; interval $[-0.00301, 0.05728]$; $q = 0.217$), negative in Mammalia (-0.00479 ; $n = \frac{25}{32}$; interval $[-0.01179, 0.00010]$; $q = 0.217$), and null within *Drosophila* ($+0.00088$; $n = \frac{13}{17}$; interval $[-0.00634, 0.01317]$; $q = 0.839$). Thus these data do not support a uniform positive within-clade pattern. Leave-one-species and leave-one-class ranges are retained in the result table; they do not hide the negative mammal or null *Drosophila* results.

Saturation shape. A centered quadratic class-fixed diagnostic gave a positive, not negative, quadratic coefficient ($+0.00226$; $n = 90$; bootstrap interval $[0.000005, 0.00510]$; sensitivity $q =$

0.0602). Saturation predicts concavity. A quadratic is not a mechanistic saturation curve, but its observed convex direction does not support the predicted GC3 saturation shape.

Recovered assembly modality. The corrected Tier 3a origin/main SweepGA run used native `--num-mappings 1:1` mappings followed by fresh IMPG queries. It produced three biological H1/H2 comparisons over deterministic H1-native coding panels ($n = 2$ eligible haplotypes per pair), not genome-wide deposited-individual heterozygosity. Coding-gene diversity, defined as H1/H2 alternative alleles per callable coding base, was 0.013852 for *Menidia menidia* (28,233/2,038,234; 1,000-replicate 50-kb block-bootstrap 95% interval [0.012354, 0.015340]), 0.00039796 for *Spinachia spinachia* (803/2,017,806; [0.00032951, 0.00048123]), and 0.00167799 for *Tautogolabrus adspersus* (3,551/2,116,227; [0.00123835, 0.00212181]). The roughly 35-fold range is heterogeneity, not an N_e trend: these are three selected assembly pairs without a common population sampling frame.

The corresponding reference-conditioned $\frac{\pi_S}{\pi_W}$ point estimates were 0.6725, 0.3169, and 0.7603. Writing variant counts as v and callable fourfold-site counts as d , they are computed as $\frac{v_S}{v_W} \frac{d_S}{d_W}$. Because paired ratio-bootstrap draws were not published, conservative intervals formed by dividing the marginal 95% bounds were respectively [0.4827, 0.9330], [0.0685, 1.7020], and [0.2780, 2.1252]; the latter two include one. This ratio is descriptive and is not polarized SFS- B .

Recovered population modality. Tier 3b analyzed two *Anopheles coluzzii* cohorts, each with 20 wild diploid individuals (40 nominal chromosomes), over the same 21-Mb region but with population-specific callable masks. Population π is the sum of unbiased pairwise differences divided by callable sites. It was 0.0123500 in AO Luanda (numerator 184,914.842 over 14,972,821 callable sites; 10,000-replicate chromosome-stratified 1-Mb block-bootstrap 95% interval [0.0118634, 0.0128173]) and 0.0158698 in GM Wali Kunda (237, $\frac{893.579}{14}$, 990,338; [0.0153949, 0.0163385]). The intervals do not overlap, documenting within-species population heterogeneity. Their reference-conditioned $\frac{\pi_S}{\pi_W}$ estimates were 0.74266 ([0.71292, 0.77088]) and 0.72356 ([0.69194, 0.75717]), using the same 10,000-replicate genomic bootstrap. Component π_S and π_W uncertainty uses a 20-individual delete-one jackknife. These population estimates are not pooled with the assembly comparisons or the single-assembly composition data. Polarized SFS- B remains deferred because the frozen plan has no outgroup, ancestral-error model, demographic model, or power threshold.

Claim boundary. Measurement denominators identify the precision target within an assembly; species-bootstrap intervals describe a different, conditional source of uncertainty. Neither identifies causality. Sparse Buffalo recombination fields do not define predeclared machinery classes, so no post-hoc recombination-class fit was invented. Composition alone cannot separate gBGC from mutation bias, selection, life history, phylogeny, or assembly/annotation structure. Taken together, the mixed class-specific directions, convex shape, modality-limited diversity recovery, and deferred SFS mean that this run **does not establish gBGC saturation or a causal gBGC explanation of the high-population diversity plateau**. It leaves the formal mechanism as a testable hypothesis rather than an empirical conclusion.

9. Limits and open questions

The saturation flattens **W/S-site** diversity; genome-wide flattening is partial:

- **W/S fraction.** Only sites that are (or recently were) W/S polymorphisms feel gBGC directly. A partial W/S fraction with an N_e -independent floor and a neutral remainder leaves a mix $\pi = (1 - X) \cdot 4N_e\mu + X \cdot (2\mu/(u\delta))$, which is flat only if $X = 1$ (Lean: [GenomeMix.lean](#)). The linked neutral reduction from gBGC at W/S sites broadens the effect.
- **Effective u .** The floor magnitude hinges on u_{eff} , uncertain by orders of magnitude and concentrated in hotspots.
- **Saturation vs. over-compression.** With the genome-wide u , γ is huge; the self-limitation and hotspot concentration are what keep the bite finite. The distribution of local u sets the genome-wide average.
- **Comparables.** Whether gBGC's N_e -scaled saturation can be distinguished from, or combined with, weak genetic draft ([Achaz & Schertzer 2023](#)) and reassessed linked selection ([Buffalo 2021](#)) is an open empirical question.

10. The formalization

The argument is machine-checked in a 16-module Lean 4 library (depending on Mathlib) at [lean/src/LewontinParadox](#). It contains, with no sorry, the marginal-coalescent invariance and the master refutation of the unbiased “homogenization” claim, the gBGC strength and its N_e -linearity, the composition equilibrium, the self-limiting saturation, the mutation–selection–drift crossover, the linkage-based distinction between repeat and single-copy diversity, the genome-mix flatness theorem, and the recombination-homology feedback. The single imported hypothesis is the Kingman mean $E[T_{\text{MRCA}}] = 2N_e$; everything else is proven. The build is reproducible:

```
cd lean && lake build
```

11. References

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